

UNIVERSITY OF BUEA

**FACULTY OF ENGINEERING
AND TECHNOLOGY**

**DEPARTMENT OF
COMPUTER ENGINEERING**

**A DISTRIBUTED BLOCKCHAIN-BASED APPROACH TO
PROTECT IDENTITY, INTEGRITY AND PRIVACY IN
AGRICULTURAL DATA PROCESSING**

By

FORCHA GLEN BELOA
(FE22P049)

M.Eng. in Telecommunications and Networks

A Thesis submitted to the Faculty of Engineering and Technology of the
University of Buea in partial fulfilment of the requirements
for the degree of Doctor of Philosophy (PhD)
in Computer Engineering.

Supervisors

Michael Ekonde SONE, PhD
Associate Professor

Elie FUTE TAGNE, PhD
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August 2026

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DEDICATION

To . . .

CERTIFICATION

UNIVERSITY OF BUEA
FACULTY OF ENGINEERING AND TECHNOLOGY

This is to certify that the work described in the thesis entitled **“A Distributed Blockchain-Based Approach to Protect Identity, Integrity and Privacy in Agricultural Data Processing”**

By _____

Was carried out in the Department of Computer Engineering

Under the supervision of

(Name and Signature of Supervisor)

Date _____

ACKNOWLEDGEMENTS

ABSTRACT

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LIST OF ABBREVIATIONS

ABAC	Attribute-Based Access Control
ACK	Acknowledgement
ADC	Analogue-to-Digital Converter
API	Application Programming Interface
ARM64	64-bit Advanced RISC Machine architecture
BATMAN-adv	Better Approach To Mobile Ad-hoc Networking, advanced
BLE	Bluetooth Low Energy
BFT	Byzantine Fault Tolerance
CA	Certificate Authority
CAP	Consistency, Availability, Partition tolerance
CPU	Central Processing Unit
CRT	Chinese Remainder Theorem
DID	Decentralised Identifier
DLT	Distributed Ledger Technology
DP	Differential Privacy
EDP	Energy-Delay Product
ESP32	Espressif 32-bit microcontroller with integrated Wi-Fi and Bluetooth
GDPR	General Data Protection Regulation
GW	Gateway
HLF	Hyperledger Fabric
HRBAC	Hierarchical Role-Based Access Control
IoT	Internet of Things
IPFS	InterPlanetary File System
JSON	JavaScript Object Notation
LoRa	Long Range (sub-GHz spread-spectrum radio modulation)
LoRaWAN	Long Range Wide Area Network
LZW	Lempel-Ziv-Welch compression
MAC	Medium Access Control
MCU	Microcontroller Unit
MSP	Membership Service Provider
PBFT	Practical Byzantine Fault Tolerance
PKI	Public Key Infrastructure
RAM	Random Access Memory
RBAC	Role-Based Access Control
RF	Radio Frequency
ROI	Return on Investment
RPi	Raspberry Pi
RTC	Real-Time Clock
SF	Spreading Factor
SHA	Secure Hash Algorithm
SLO	Service Level Objective
SSS	Shamir Secret Sharing

TEE	Trusted Execution Environment
TPS	Transactions Per Second
TPM	Trusted Platform Module
ZKP	Zero-Knowledge Proof

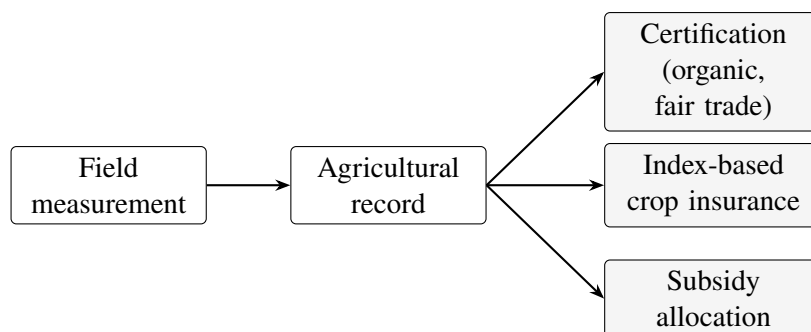
CHAPTER ONE

INTRODUCTION

1.1. Background to the Study

Agriculture is undergoing a transition from experience-driven practice to data-driven decision making. Networks of low-cost sensors now measure soil moisture, canopy temperature, humidity and nutrient status continuously, and the resulting streams feed irrigation controllers, yield models and certification schemes [1, 2]. In water-stressed regions this shift is not a convenience but a necessity. Irrigation accounts for the dominant share of freshwater withdrawal, and climate variability has compressed the margin for error in scheduling decisions [3, 4].

A measurement taken in a field does not stop at the field. It becomes a record, and that record is consumed by parties with interests of their own, as Figure 1 sets out. The value of the record to those parties is exactly what creates the incentive to falsify it, and the further a record travels from the instrument that produced it, the weaker the evidence that it is genuine.



*Each downstream use attaches value to the record,
and therefore creates an incentive to falsify it.*

Figure 1. Lifecycle of an agricultural measurement. A reading becomes a record that is consumed by parties outside the farm, and the value those parties attach to it is what makes falsification worthwhile.

The economics of the sensing layer explain why the problem is hard rather than

merely tedious. A deployment that is worth installing on a smallholding must cost a small fraction of the crop value it protects, which forces the node bill of materials down to a level at which neither a hardware security module nor a generous energy budget is affordable. The same economics rule out mains power and wired backhaul across a field, so the design space collapses onto battery-powered microcontrollers speaking a long-range, low-rate radio. Every protection mechanism proposed in this thesis has to survive inside that envelope, and a mechanism that works only on a mains-powered gateway has solved the easier half of the problem.

The sensing layer of such systems is built from severely constrained devices. An Espressif 32-bit microcontroller with integrated Wi-Fi and Bluetooth (ESP32) node operating on battery and harvested solar power must transmit over kilometre-scale distances using Long Range (sub-GHz spread-spectrum radio modulation) (LoRa), a modulation scheme that trades data rate for range and therefore charges a high airtime cost per packet [5, 6]. Airtime is the binding constraint. It determines energy consumption at the node, and, through duty-cycle regulation and collision probability, it determines how many nodes a single gateway can serve [7]. Conventional responses compress the payload, but general-purpose compressors such as Huffman and Lempel-Ziv-Welch compression (LZW) carry dictionary and buffer overheads that a microcontroller with a few kilobytes of free Random Access Memory (RAM) can ill afford [8, 9].

Two further constraints sharpen this. Sub-GHz Internet of Things (IoT) bands are governed by a duty-cycle limit, so a node that has just transmitted must stay silent for a period proportional to the airtime it consumed. Airtime is therefore not only an energy cost but a rate limit: it caps how often a node may report, and no increase in battery capacity relaxes it. Second, because LoRa uses unslotted access, packets collide whenever two nodes transmit on the same channel at the same time, and the collision probability

grows with the square of offered traffic in the regime that matters. Halving per-packet occupancy therefore buys more than a proportional reduction in collisions, which is why the mechanism studied here is worth the arithmetic it costs.

The Chinese Remainder Theorem (CRT) offers a different route. Decomposing a reading into residues modulo pairwise coprime moduli yields several short packets in place of one long one [10, 11]. Because a multi-channel concentrator such as the SX1302 demodulates channels concurrently, the short residue packets can be received simultaneously, and the original value is recovered in $O(k)$ operations for k moduli. The practical effect is a reduction in per-packet occupancy rather than in total transmitted volume, which is precisely the quantity that governs gateway capacity. The detailed encoding and reconstruction path is presented in Chapter Three.

Residue arithmetic is not new to distributed systems. The same decomposition has been applied to the throughput problem in permissionless blockchains, where splitting work across residue classes permits parallel processing of operations that would otherwise serialise. That line of work establishes the arithmetic and the parallelisation argument; what it does not address is the constrained radio link, where the payoff is airtime rather than processor cycles, nor the identity and confidentiality questions that arise when the residues cross a shared medium. This thesis takes the arithmetic as established and asks what it is worth on the edge of the network rather than at its centre.

Data that has been collected cheaply must then be trusted. Agricultural records increasingly carry consequences beyond the farm gate. They substantiate organic and fair-trade certification, they underwrite index-based crop insurance, and they inform subsidy allocation. Each of these uses creates an incentive to falsify. A centralised database concentrates that risk in a single administrator, and an unprotected sensor network invites the injection of fabricated readings by any party with a compatible radio. Distributed

Ledger Technology (DLT) addresses the tamper-evidence requirement by binding records into a hash-linked chain replicated across mutually distrusting parties, and permissioned platforms such as Hyperledger Fabric (HLF) add an identity layer through a Membership Service Provider (MSP) and Public Key Infrastructure (PKI), making it feasible to say not merely that a record is unaltered but who produced it [12–14].

Running that machinery at the edge is the difficulty. Consensus, endorsement and world-state maintenance were designed for datacentre hardware, whereas an agricultural deployment must place gateways in the field, on 64-bit Advanced RISC Machine architecture (ARM64) single-board computers, behind intermittent connectivity. Whether a permissioned ledger can meet irrigation-control timing requirements on such hardware has until recently been asserted rather than measured.

Finally, protection is not exhausted by integrity. A soil-moisture trace is also a disclosure. It reveals plot productivity, irrigation strategy and, in aggregate, the commercial position of the farmer. Where records are replicated across a consortium that includes buyers, insurers and cooperatives, the same replication that secures integrity enlarges the audience for private information. Identity, integrity and privacy therefore have to be treated as one design problem rather than three independent ones, which is the position this thesis takes.

1.2. Statement of the Problem

Three properties that ought to hold simultaneously are, in current systems, traded against one another.

Identity is weak at the sensing layer. Constrained nodes seldom carry credentials strong enough to authenticate individual readings. Where authentication exists it typically terminates at the gateway, so the ledger attests to what the gateway submitted rather than to what the sensor measured. The provenance chain has a gap exactly where falsification

is cheapest, at the final hop between an individual node and the record that claims to represent it.

Integrity guarantees do not survive the move to the edge. Permissioned ledgers provide tamper evidence, but their published performance characterisations assume server-class hosts. On ARM64 gateways with constrained memory and intermittent links, it is not established that endorsement and ordering can complete inside the control loop of an irrigation decision, nor how the ledger behaves during network partition and recovery.

Privacy is sacrificed to obtain integrity. Replicating raw agronomic readings to every consortium member is the simplest way to make them verifiable and the worst way to keep them confidential. Existing deployments largely accept this trade, or push confidential data off chain in a way that reintroduces the trusted intermediary the ledger was meant to remove.

The three shortfalls, shown together in Figure 2, are not independent faults to be repaired one at a time. Each follows from the same scarcity, and a remedy applied to one is paid for out of the budget available to the others.

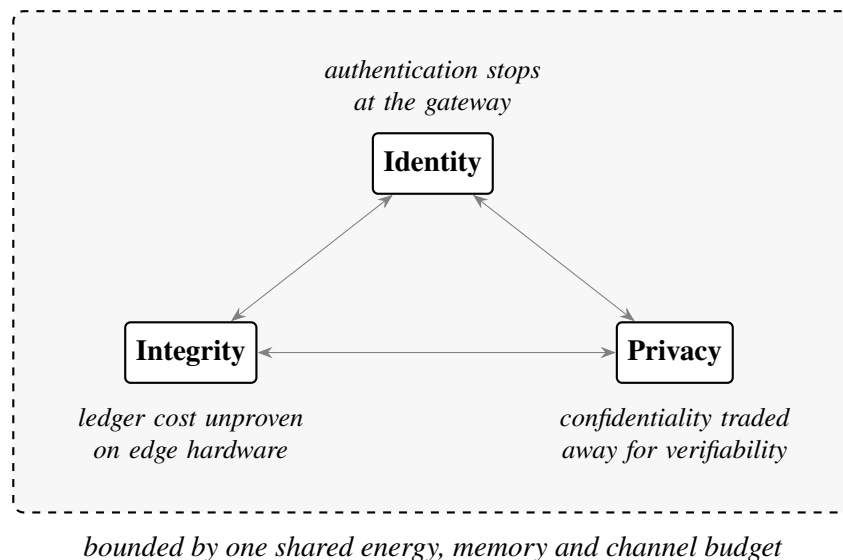


Figure 2. The three protection properties and the shortfall in each. Because all three draw on a single constrained budget, they form one design problem rather than three independent ones.

Underlying all three is a resource problem. The transmission budget that would pay for per-reading signatures, encryption and privacy-preserving encodings is already consumed by the raw payload itself. Any scheme that adds cryptographic protection without first reducing transmission cost will exhaust the node's energy budget or the gateway's channel capacity.

How, then, can identity, integrity and privacy be protected simultaneously in an agricultural data processing pipeline, end to end from constrained sensing node to distributed ledger, within the energy, memory and channel-capacity budgets that field deployment actually permits?

1.3. Rationale of the Study

The budget for protection has to come from somewhere. If CRT residue decomposition reduces per-packet airtime substantially, it frees channel capacity and node energy that can be reinvested in protection rather than merely banked as efficiency. The same decomposition has a second, less obvious property. An individual residue discloses little about the original value, so residues distributed across channels or custodians provide a natural substrate for confidentiality and for threshold reconstruction. A mechanism introduced for efficiency can thus be made to serve privacy, which is what makes the combination worth studying rather than treating transmission and security as separate layers.

The two mechanisms also compose cleanly. Residue decomposition and ledger verification operate on the same object at different points in its life, the first on the reading in flight and the second on the record at rest, and the quantity each is sensitive to is different: the radio cares about occupancy, the ledger about the size and count of committed transactions. A scheme that reduces occupancy without increasing transaction count therefore improves one without taxing the other, which is not generally true of

protection mechanisms and is worth demonstrating rather than assuming.

There is also an evidential gap. Claims about edge blockchain feasibility are typically supported by simulation or short laboratory runs. A sustained field deployment on real hardware, through real weather and real connectivity failures, produces a different and more useful class of evidence, including the failure modes that simulation does not generate.

The applied case is direct. Smallholder and arid-region agriculture is where both the water-efficiency gains and the certification-fraud exposure are largest, and where infrastructure assumptions of existing systems are least likely to hold.

1.4. Objectives of the Study

1.4.1. Main Objective

To design, implement and evaluate a distributed blockchain-based framework that protects identity, integrity and privacy in agricultural data processing, and to validate it in a sustained field deployment on constrained edge hardware.

1.4.2. Specific Objectives

1. **Analyse** the identity, integrity and privacy requirements of agricultural data pipelines, decomposing the threat surface from sensing node through gateway to ledger into an attacker model with stated capabilities and assumptions.
2. **Design** an integrated protection scheme combining CRT multi-channel residue transmission, which lowers per-packet airtime while preserving exact reconstruction, with an identity layer that binds each reading to an authenticated originating node through HLF MSP credentials and Hierarchical Role-Based Access Control

(HRBAC) policy enforced in chaincode.

3. **Implement** the framework on a heterogeneous edge cluster of ESP32 sensing nodes and ARM64 Raspberry Pi (RPi) gateways running HLF under Raft ordering, deployed under real agricultural operating conditions.
4. **Evaluate** the transmission, ledger and access-control layers against compression and centralised baselines on airtime, energy, throughput, latency and memory footprint, and assess the scalability and fault-tolerance envelope through queuing-theoretic analysis validated against measured load.

1.5. Research Questions

- RQ1.** To what extent can CRT residue decomposition reduce per-packet airtime and increase gateway packet capacity without loss of reconstruction fidelity?
- RQ2.** Can a permissioned distributed ledger sustain the throughput and latency required by real-time irrigation control when hosted entirely on ARM64 edge gateways?
- RQ3.** What identity binding is required for a ledger record to attest to the originating sensing node rather than merely to the submitting gateway, and what does that binding cost?
- RQ4.** What confidentiality is obtained by distributing residues across channels and custodians, and under what adversary model does it hold?
- RQ5.** How does the integrated framework behave as offered load approaches gateway and ledger saturation, and where are its practical limits?

1.6. Research Hypotheses

H1 (Transmission efficiency). CRT residue transmission reduces per-packet airtime by at least 40% relative to raw transmission while maintaining at least 99% reconstruction accuracy.

H2 (Edge ledger performance). HLF hosted on four RPi 4B gateways sustains a mean transaction latency below 2 s at an average throughput of at least 50 TPS.

H3 (Access-control overhead). Enforcing Role-Based Access Control (RBAC) in chaincode preserves at least 50 TPS and keeps mean smart-contract execution below 500 ms.

H4 (Field robustness). Sustained field operation maintains at least 99% end-to-end reliability, with measured airtime within 5 ms of the analytical model.

H5 (Privacy preservation). An adversary observing fewer than k of the k residue channels cannot reconstruct the original reading, and the residual information leakage remains below a stated bound.

1.7. Scope and Delimitations

In scope. Sub-GHz LoRa sensing networks of the order of fifty nodes; permissioned ledgers of the HLF family under Raft ordering; gateway hardware in the RPi 4B class; agronomic telemetry, specifically soil moisture, temperature, humidity and irrigation actuation; and the identity, integrity and privacy properties of that pipeline.

Out of scope. Public permissionless blockchains and their consensus economics; cellular and satellite backhaul; image-based and remote-sensing agronomy; agronomic model accuracy as such, which is treated as given rather than as an object of study; and formal cryptographic proofs of the underlying primitives, which are cited rather than

re-derived.

Delimitations. Field validation is drawn from a single deployment site and a single cropping season, which bounds the claims that can be made about generalisation across soil types and climates. Node population is bounded by what could be instrumented. Behaviour beyond that population is established analytically and by synthetic load rather than by direct measurement.

1.8. Significance of the Study

For **engineering practice**, the work supplies measured rather than simulated evidence about what a permissioned ledger costs on edge hardware, and a transmission scheme that reduces that cost enough to make protection affordable.

For **the research literature**, it connects two bodies of work that have developed separately, namely residue arithmetic for communication efficiency and distributed ledgers for data provenance, and shows that the first can subsidise the second.

Methodologically, the work contributes a characterisation obtained under field conditions rather than in a laboratory. Failures that a testbed does not generate, mesh partitions, thermal derating, intermittent backhaul and the recovery behaviour that follows each, are exactly the ones that determine whether such a system is deployable, and they only appear when the system is left running.

For **agricultural stakeholders**, it offers a path to verifiable certification and insurance records that does not require farmers to surrender commercially sensitive detail to every consortium member.

1.9. Definition of Key Terms

Airtime The time a radio occupies a channel to transmit one packet. It governs both node energy consumption and gateway capacity.

Residue The remainder of a reading modulo one of a set of pairwise coprime moduli. A complete residue set uniquely determines the original value below the product of the moduli.

Integrity The property that a record has not been altered since it was committed, and that alteration would be detectable.

Identity The property that a record can be attributed to a specific authenticated principal, here a specific sensing node rather than merely a gateway.

Privacy The property that a record's content is disclosed only to principals authorised to see it, notwithstanding replication of the ledger.

Endorsement In HLF, the process by which designated peers simulate a transaction and sign the result before ordering.

1.10. Organisation of the Thesis

Chapter One states the background, problem, rationale, objectives, questions and hypotheses. **Chapter Two** reviews the literature on precision agriculture and IoT sensing, transmission optimisation, distributed ledgers in agriculture, edge-resident consensus and privacy-preserving data processing, and identifies the gap this work addresses. **Chapter Three** presents the materials and methods: system architecture, CRT formulation, identity and access-control design, hardware implementation, deployment protocol and evaluation methodology. **Chapter Four** reports results and applications across the transmission, ledger, access-control and agricultural-outcome dimensions. **Chapter Five** discusses the findings against the hypotheses, states limitations and threats to validity, and sets out conclusions, recommendations and perspectives for further research.

CHAPTER TWO REVIEW OF THE LITERATURE

2.1. Introduction

This chapter surveys the five bodies of work on which the thesis draws: precision agriculture and IoT sensing; low-power wide-area transmission and the airtime constraint that governs it; residue arithmetic and the CRT; distributed ledgers in agriculture and their migration to the network edge; and identity, access control and privacy for replicated records. Each is treated on its own terms, then criticised, and the chapter closes by naming what the five leave undone.

2.2. Precision Agriculture and IoT Sensing

2.2.1. From Wireless Sensor Networks to Managed Irrigation

Precision agriculture began as site-specific crop management, matching inputs to measured within-field variability rather than to field averages [15, 16]. Early instrumentation used wireless sensor networks for soil and microclimate monitoring [17, 18], and the subsequent decade moved the field from measurement to actuation, with Long Range Wide Area Network (LoRaWAN) irrigation control among the clearest applications [19]. Reviews of the area agree on the technical progress and on a persistent weakness [20, 21]: integration between the sensing layer and the decision-support systems that consume it remains fragmented, so measurement quality is rarely the limiting factor on outcomes.

2.2.2. What the Sensing Literature Does Not Settle

Two questions are left open. First, almost all of this work treats the reading as trustworthy by construction: the sensor reports, the system believes it, and no mechanism attests that

the value committed to storage is the value measured. Second, the economics that make a deployment viable on a smallholding, which force the node bill of materials to a few tens of dollars, are seldom carried through to the security design. A protection mechanism that assumes a secure element or a mains supply has assumed away the constraint that defines the problem.

2.3. Low-Power Wide-Area Transmission and the Airtime Constraint

2.3.1. The Airtime Model

LoRa is the dominant physical layer for agricultural telemetry, with demonstrated rural ranges of several kilometres and, in favourable terrain, up to 15 km [6, 22, 23]. Its cost is airtime. For a spreading factor SF , bandwidth BW and coding rate CR , the symbol duration and the time on air of a packet carrying PL payload bytes are

$$T_{\text{sym}} = \frac{2^{SF}}{BW}, \quad (2.1)$$

$$T_{\text{air}} = T_{\text{preamble}} + T_{\text{sym}} \left(8 + \max \left(\left\lceil \frac{8PL - 4SF + 28 + 16}{4(SF - 2DE)} \right\rceil (CR + 4), 0 \right) \right), \quad (2.2)$$

where DE is the low-data-rate optimisation flag. The term that matters is the ceiling: airtime grows in discrete steps with payload length, and every additional symbol is charged at T_{sym} , which doubles with each increment of SF [6, 24].

Airtime is not merely an energy cost. Sub-GHz bands impose a duty-cycle limit, so a node that has transmitted for T_{air} must remain silent for a proportional interval, which caps reporting frequency independently of battery capacity. Under the unslotted access that LoRaWAN class A uses, the offered load per channel over a reporting window T_w for

N nodes is

$$G = \frac{N \cdot T_{\text{air}}}{T_w}, \quad (2.3)$$

and the pure-ALOHA collision probability follows as

$$P_{\text{coll}} = 1 - e^{-2G} \approx 2G \quad \text{for small } G, \quad (2.4)$$

which is linear in T_{air} in the low-load regime and degrades sharply as G approaches unity [6, 24]. Reducing per-packet airtime therefore buys capacity twice over, once through the duty cycle and once through the collision term.

2.3.2. Compression and Sampling Responses

The conventional response is to shorten the payload. Lossless coders adapted to sensor streams, principally LZW and Huffman variants, achieve useful ratios on agronomic data [8, 25], but their dictionary and buffer requirements are measured in kilobytes, which is a significant fraction of the free RAM on a microcontroller already running a radio stack and a sensor driver [9]. Adaptive sampling transmits only when a reading departs from a predicted trajectory [26], at the risk of suppressing exactly the transient that mattered. Aggregation and protocol-level optimisation [27, 28] reduce traffic at the cost of resolution or latency.

2.3.3. The Common Limitation

Every one of these approaches reduces the number of bytes sent. None reduces the occupancy of an individual packet while preserving the reading exactly, and occupancy is

what Equations 2.3 and 2.4 are sensitive to. This distinction, between total volume and per-packet occupancy, is where the present work departs from the compression literature.

2.4. Residue Arithmetic and the Chinese Remainder Theorem

2.4.1. Statement and Reconstruction

Let $m_1, m_2, \dots, m_k$ be pairwise coprime moduli and $M = \prod_{i=1}^k m_i$. For any integer x with $0 \leq x < M$, the residue vector

$$\mathbf{r} = (r_1, r_2, \dots, r_k), \quad r_i = x \bmod m_i, \quad (2.5)$$

determines x uniquely, and x is recovered by

$$x = \left(\sum_{i=1}^k r_i M_i N_i \right) \bmod M, \quad M_i = \frac{M}{m_i}, \quad N_i = M_i^{-1} \bmod m_i. \quad (2.6)$$

Reconstruction is exact because $M_i N_i \equiv 1 \pmod{m_i}$ and $M_i \equiv 0 \pmod{m_j}$ for $j \neq i$ [11, 29]. Evaluated by Garner's algorithm the cost is $O(k)$ modular operations with no division in the inner loop [10], which is what makes the scheme viable on a gateway rather than only on a workstation. Figure 3 shows the decomposition and its inverse.

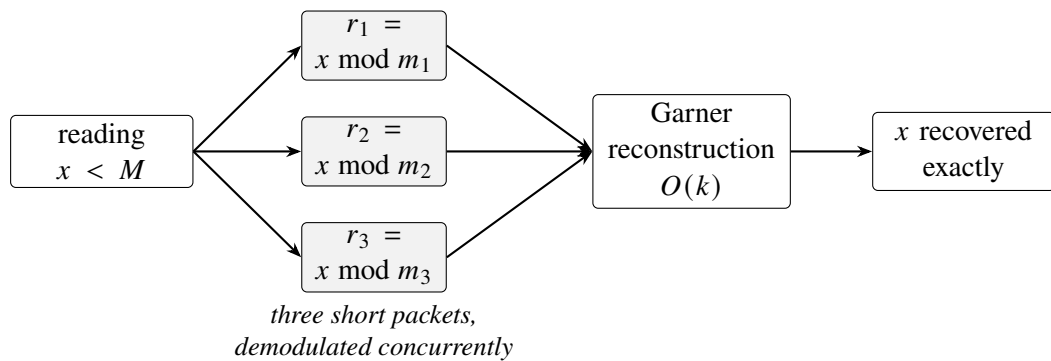


Figure 3. Residue decomposition and reconstruction. One long packet is replaced by k short ones that a multi-channel concentrator can demodulate simultaneously, and the original value is recovered exactly in $O(k)$ operations.

2.4.2. Where Residue Systems Have Been Applied

Residue number systems built on Equation 2.6 are long established in computer arithmetic, where carry-free addition and multiplication across residue channels accelerate arithmetic and support fault-tolerant computation [30, 31]. The theorem is equally standard in cryptography [10, 11] and in error correction [32]. Within sensor networks, Wang, Chen, and Wang [33] applied residue representations to reduce stored data volume.

2.4.3. The Gap in the Residue Literature

The sensor-network application targeted storage, not transmission, and therefore did not exploit the property that matters here: that k residues are k independent short packets which a multi-channel concentrator can receive at the same instant. The computer-arithmetic literature exploits parallelism, but across arithmetic units rather than across radio channels, where the currency is occupancy rather than processor cycles. No prior system, to the author's knowledge, uses residue decomposition to drive parallel demodulation on a low-power radio.

2.5. Distributed Ledgers in Agriculture

2.5.1. Supply-Chain Transparency

The earliest and best-developed application is provenance along the food supply chain. Tian [34] combined RFID with a blockchain to trace agri-food products end to end, and subsequent surveys map the field [35, 36] and its food-safety consequences [37]. The common architecture places the ledger between organisations that have commercial reason to distrust one another, which is precisely the setting in which replicated tamper-evident storage earns its cost.

2.5.2. Farm Data Management and Smart-Contract Automation

A second line applies the ledger inside the farm, to data ownership and access rather than to product movement [38–40]. A third automates agreements, using smart contracts to trigger payments, insurance settlement or irrigation actions [41, 42]. Both assume a ledger that is reachable and responsive.

2.5.3. The Recurring Weakness

Across all three lines, implementations that operate under the constraints of real-time control on resource-constrained hardware remain scarce. Where performance is reported it is typically obtained on server-class hosts, and the question of whether endorsement and ordering complete inside an irrigation control loop is not usually asked.

2.6. Edge-Resident Consensus and Constrained Hardware

2.6.1. Edge Computing in Agricultural Systems

Moving computation to the field reduces round-trip latency and reduces dependence on backhaul that rural sites often lack [43]. Liu et al. [44] reported improved response times for irrigation control under edge processing, and Wang et al. [45] showed that keeping data local improves privacy by limiting what leaves the site. ARM64 single-board clusters are now routinely used for distributed sensing and inference [43, 45].

2.6.2. Networking the Edge

Agricultural deployments overwhelmingly use star topologies, in which a single gateway failure removes a zone. Mesh routing has been studied mostly in urban and industrial

settings [46]; BATMAN-adv has shown promise for rural connectivity [47] without being taken up in precision agriculture. Hybrid multi-technology architectures have been proposed [48] but tend to add management complexity without integrating cleanly.

2.6.3. Capacity Limits

Where a gateway serves C channels over a window W and each node consumes T_{eff} of effective airtime per reporting cycle, the node ceiling is

$$N_{\text{max}} = \frac{C \cdot W}{T_{\text{eff}}}, \quad (2.7)$$

and per-channel packet throughput, for a concentrator that demodulates one packet per channel concurrently, is

$$\lambda = \frac{1}{T_{\text{air}}}. \quad (2.8)$$

Equations 2.7 and 2.8 pull in different directions once a reading is split across channels: splitting raises the total airtime a node consumes, and so lowers N_{max} , while lowering T_{air} per packet, and so raising λ . Which of the two binds depends on whether the deployment is limited by node count or by packet capacity, a distinction the existing literature does not draw.

2.6.4. What Is Missing at the Edge

Edge implementations rarely integrate with verifiable storage, so the trust gap that motivates the ledger reappears the moment computation is moved off the ledger. Communication parallelism, meaning simultaneous use of multiple wireless channels

by one logical reading, is essentially unexplored.

2.7. Identity, Access Control and Privacy

2.7.1. Identity and Access Control

Permissioned ledgers supply identity through a MSP backed by a PKI, which allows a record to be attributed to an enrolled principal rather than merely to an address [12, 13]. RBAC and its hierarchical variants are the usual policy layer, enforced in chaincode so that the authorisation decision is itself auditable [39].

2.7.2. Privacy Under Replication

Replication is what makes a ledger tamper-evident and is also what makes it leak. Where a consortium includes buyers, insurers and cooperatives, every member sees every replicated record, and agronomic traces disclose plot productivity and irrigation strategy. Keeping data local reduces the exposure but reintroduces the trusted party the ledger was meant to remove [45].

2.7.3. The Unresolved Tension

The literature offers identity and integrity together, and privacy separately, usually by withholding data from the ledger. What it does not offer is a mechanism in which the same operation that makes transmission affordable also constrains disclosure. That possibility, latent in the residue representation of Equation 2.5, is the opening this thesis works in.

2.8. Research Gaps

Table 2 states the gaps that follow from the five preceding sections and the objective that addresses each.

Table 2. Gaps identified in the literature and the objective addressing each.

Area	Gap	Objective
Transmission cost	Compression reduces total volume, not per-packet occupancy, and costs memory a constrained node does not have	O2
Residue arithmetic	Applied to storage and to arithmetic units, never to parallel demodulation on a low-power radio	O2
Edge ledgers	Performance characterised on server hardware; behaviour on ARM64 gateways under partition unmeasured	O3, O4
Identity	Authentication terminates at the gateway, so the record attests to the submitter rather than the sensor	O1, O2
Privacy	Confidentiality obtained by withholding data, which reinstates a trusted intermediary	O1, O2
Capacity	No validated model distinguishing node-count from packet-capacity limits for multi-channel residue traffic	O4

2.9. Partial Conclusion

The sensing literature establishes what to measure and treats the measurement as trustworthy. The transmission literature establishes that airtime binds and attacks it by shortening payloads, which does not address per-packet occupancy. Residue arithmetic supplies a decomposition whose parallelism has been exploited in silicon but not on the air. The ledger literature supplies tamper evidence and identity, characterised on hardware that agricultural deployments do not have. The privacy literature obtains confidentiality by withholding, which forfeits verifiability.

The four therefore leave a single composite question, which the remainder of this thesis addresses: whether a decomposition adopted to reduce per-packet occupancy can

simultaneously carry node-level identity and bound disclosure, and whether the resulting system holds up on the hardware and in the conditions that field deployment actually imposes.

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APPENDIX A

SUPPLEMENTARY MATERIAL